

W.E.B. Du Bois, Problem Framing, and the Politics of Abstraction

“Between me and the other world, there is ever an unasked question: How does it feel to be a problem?”

– W.E.B. Du Bois

Problem Statement

- ▶ Machine learning and quantitative measurement rely on assumptions, normative practices, and values that often go underexamined. These latent aspects of ML and quantitative practice are treated as technical defaults rather than historical or political choices.
- ▶ Abstraction and problem framing can formalize social life in ways that obscure history, context, and harm.
- ▶ This project asks what happens when we refuse to treat these abstractions as neutral.

Comte, Galton, and the History of Social Statistics

- ▶ Comte helped establish positivist social science and an abstract object called “society,” and formalized the study of the normal vs. the pathological. (Hacking, 1990)
- ▶ Galton developed statistical methods and eugenics in his pursuit of subjecting all of life to measure and quantification (Koopman, 2019)
- ▶ Du Bois’s critique of Comte and eugenics show how abstraction can replace the study of living people with the study of “Society” as an abstract object. Social statistics and measurement often function as tools of pathologization of those populations considered to be deviant or disordered.

Du Bois and Hesitance

- ▶ In *Sociology Hesitant* (2000), Du Bois critiques the quantification and social science practice of his day.
- ▶ Hesitance is a methodological posture of skepticism, reflection, refusal, and imagination. Through Du Bois’ tables, data portraits, charts, and graphs we see his use of measurement and quantification to document the dynamics of Black life (2018; 1996).
- ▶ Here, hesitance becomes a framework for critically examining quantification and measurement

Key Fields

Alongside the work of Du Bois, this project is informed by literature across Critical Quantitative and Computational Methods (CritQCM), QuantCrit, Critical Data Studies, and Black Studies.

Methodological Implications

Hesitance is a disciplined refusal to mistake abstraction for the world. It is also a method for evaluating the assumptions, categories, and consequences of quantitative inquiry.

Hesitance Questions Across the ML Pipeline

Problem framing

Who is the object of problem framing, and who defines the problem?

Measurement

What absences, categories, and power relations are encoded in the data sets?

Modeling

What is forced to be commensurable, and what becomes noise?

Evaluation

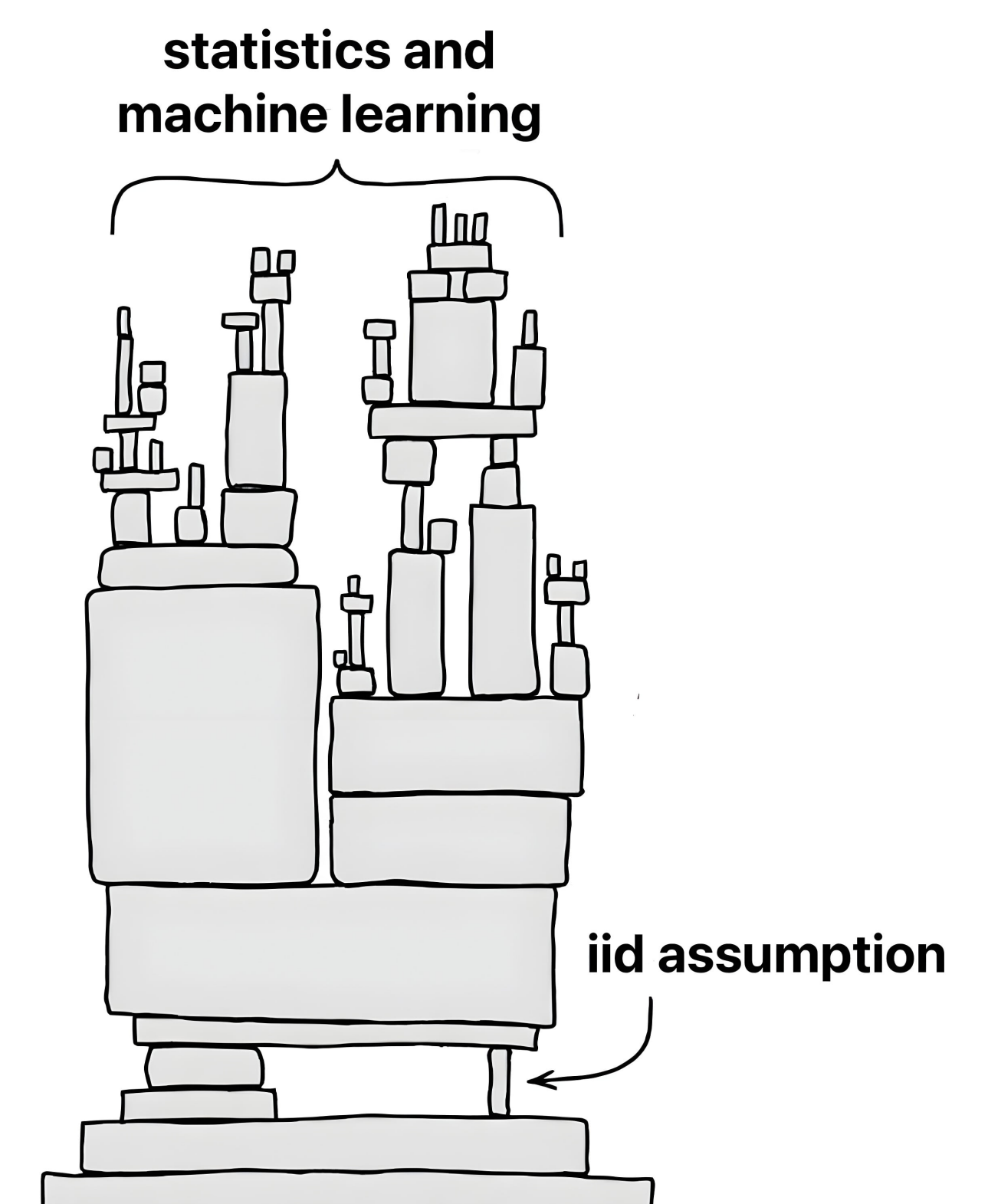
Which harms remain invisible when success is reduced to a metric?

Deployment

Who bears the burden of model error, refusal, or exclusion?

Intervention – Critically examining the i.i.d. Assumption

- ▶ The i.i.d. assumption is not only technical; it organizes what counts as legible and knowable.
- ▶ When taken for granted, it can flatten social complexity and hide epistemic commitments.
- ▶ Du Boisian hesitance offers a framework for interrogating this abstraction.



What This Project Contributes

- ▶ A Du Boisian lens for evaluating machine learning assumptions.
- ▶ A method for linking technical choices to social and historical context.
- ▶ A framework for critical quantitative practice that treats abstraction as contested.

References

For more information and a full bibliography please visit

